Zihao Zhang

School of Cyber Science and Technology, USTC
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Education

University of Science and Technology of China Hefei, China
B.E. in Information Security Sep 2022 - Present

- Overall GPA: 3.80/4.3 Rank:**4th**/73(**5.6%**)

Research Interests

Cryptography, Applied Cryptography, Zero-Knowledge Proofs

Publications

[2] **Efficient SNARK for Floating-Point Computation**
Submitted to IEEE S&P 2026
Wenjie Qu*, **Zihao Zhang*(co-first author)**, Jiaheng Zhang

[1] **Certus, Efficient Verifiable Convolutional Neural Network Inference based on Zero-Knowledge Proofs**
Submitted to Eurocrypt 2026
Wenjie Qu, **Zihao Zhang**, Yanpei Guo, Jiaheng Zhang

Research Experience

Efficient Zero Knowledge Proofs for Convolutional Neural Networks Mar 2025 – July 2025
Research Assistant at NUS Advisor: Prof. Jiaheng Zhang, Dr. Wenjie Qu

- Participated in deriving efficient prover-friendly constraints for checking multi-channel 2D convolutions in CNN by making use of relationship between matrix convolution and mathematical 2D convolution.
- Completed soundness analysis of the protocol.
- Implemented the framework using C++.
- Tested the protocol on VGG-16 and ResNet-50 and achieved obvious speedup over prior works, such as Modular (CCS'23), zkCNN (CCS'21) and zkNN (eprint'24).
- Contributed to writing and editing of the manuscript.

Efficient Zero Knowledge Proofs for Floating-Point Computations Aug 2025 – Nov 2025
Research Assistant at NUS Advisor: Prof. Jiaheng Zhang, Dr. Wenjie Qu

- Designed a mapping for unifying different numbers in the IEEE-754 standard in the circuit to make them more circuit-friendly.
- Completed analysis of edge cases for addition, multiplication and division.
- Derived decomposing scheme of exp, ln, sin and cos by leveraging their mathematical structures, which was combined with per-chunk lookup scheme to obtain an efficient ZKP framework for proving these transcendental functions.
- Implemented Spartan, Ligero ,LogUp and our SNARK in C++, and tested our SNARK.
- Contributed to writing and editing of the manuscript.

Academic Projects

Tiny-C Compiler (Course Project for Compiler Theory B) Sep 2024 – Dec 2024
Advisor: Prof. Weihai Li

- Developed a simplified C compiler in C, implementing lexical analysis, syntax analysis, and Intermediate Representation (IR) generation for the Tiny-C language.

Teaching Assistant Experience

Computer Programming Sep 2024 – Jan 2025
Advisor: Prof. Jian Tang

Selected Awards and Honors

- Outstanding Student Scholarship, **Golden Award (top 5%)** Oct 2023
- Wang Xiaomo **Talent Program** in Cyber Science and Technology Aug 2023 – July 2024
- Scholarship of Wang Xiaomo Talent Program Jan 2024
- Huawei Scholarship (**1 winner among 73 students**) Oct 2024

Relevant Courses

Computer Science: C Programming, Data Structures and Algorithms, Operating Systems, Computer Networks, Algorithms Analysis and Design, Cryptography, Compiler Theory
Mathematics: Mathematical Analysis, Linear Algebra, Foundation of Algebra, Probability Theory and Mathematical Statistics, Information Theory

Skills

Programming Languages: C/C++, Rust, Python, HTML, CSS, JavaScript
Miscellaneous: SQL, Linux, Git, LaTeX, Markdown, Torch
English: TOEFL: 106 (Reading: 29; Listening: 24; Speaking: 25; Writing: 28)

Extracurricular Activities & Interests

Member of Student Union of School of Cyber Science and Technology Sep 2023 – Jun 2024

- Organized a campus-wide badminton tournament as the person in charge. I was responsible for all tasks except financial reimbursement and event promotion

Independent Server Administration & Configuration Experience Dec 2024 – Aug 2025

- Deployed and configured a stable Ubuntu 20.04 LTS environment on a Raspberry Pi 4B independently from scratch, and built a Forge-based Minecraft server and installed mods and plugins such as Create and LuckPerms.
- Implemented a reliable NAT traversal setup using FRP to enable secure remote access.